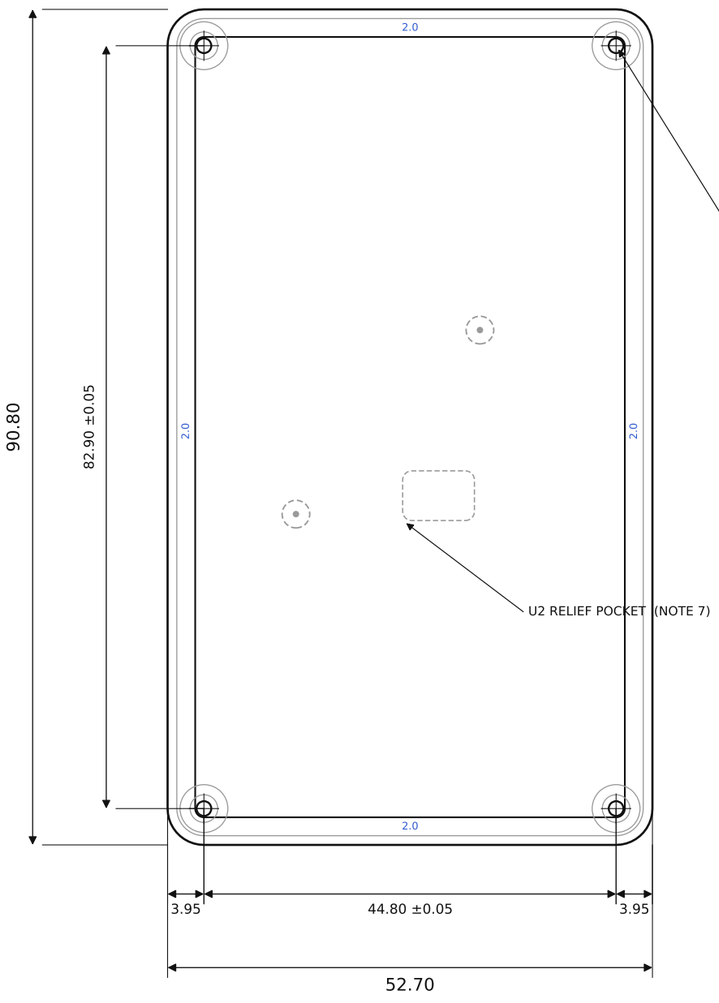
BACK FACE (outer) SCALE 1.5:1



POS	Ø0.10 (M)	A	B

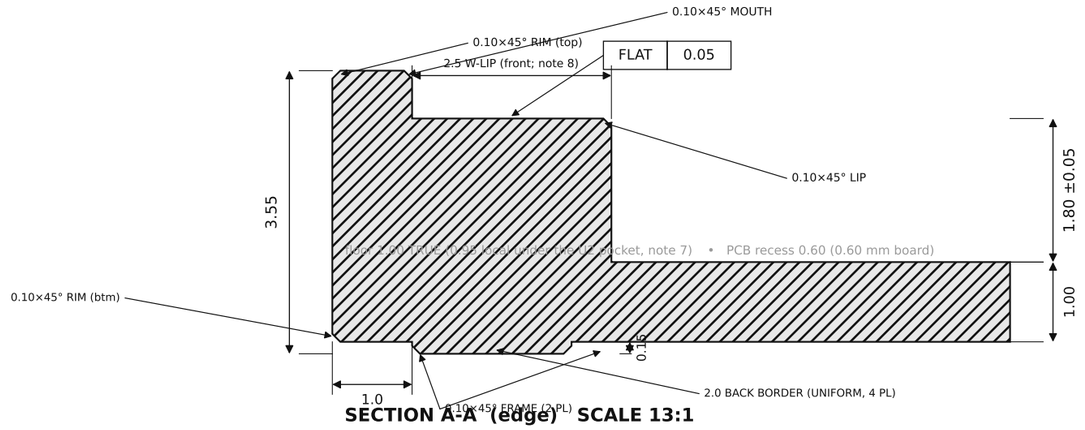
4× M2 THREAD, THRU

Ø1.6 TAP DRILL • TAP FROM BACK FACE

Ø3.0 SPOTFACE, BACK (depth per Detail B)

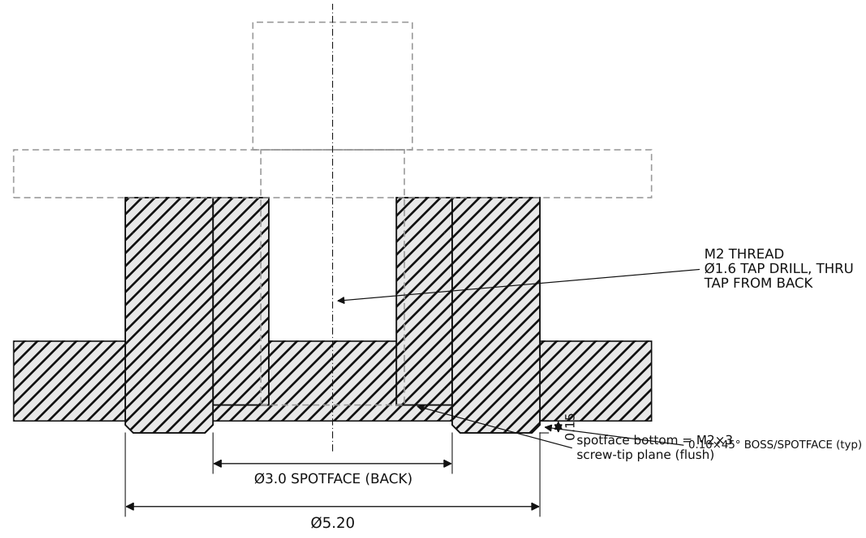
C3 = hole-pattern pitch ±0.05   C4 = M2 thread

SECTION A-A → edge profile   DETAIL B → corner boss



SECTION A-A (edge) SCALE 13:1

C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)
PCB + M2×3 BRASS SCREW — REF, NOT THIS PART



DETAIL B (corner boss) 13:1

GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)

LINEAR & HOLE-POSITION DIMS	**ISO 2768-m**
CAVITY DEPTH C1 (SECTION A-A)	**1.80 ±0.05**
HOLE-PATTERN PITCH C3 (PLAN)	**44.80 / 82.90 ±0.05**
PCB-REST FLATNESS C2	**FLAT 0.05**

SOLAR-GLOW DRH v3.0 — Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)

DWG solar-glow-drh-v3_0-backshell-0p6b-brace	REV v3.0
MATERIAL TI GR5 (TC4)	FINISH BEAD-BLAST
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1

CRITICAL DIMENSIONS — C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

C1	Cavity depth (boss-top plane → cavity floor)	1.80 ±0.05
C2	PCB-rest plane flatness (lip + 4 corner bosses)	FLAT 0.05
C3	4× mounting-hole pattern, pitch 44.80×82.90	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5.
2. FINISH: BEAD-BLAST MATTE (UNIFORM ON THE STEPPED BACK FACE). REAR ART LASER-MARKED IN THE RECESSED BACK FIELD.
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN H-BRACE REGISTERS TO THIS SHELL BY FITMENT - ITS 4 OUTBOARD RAILS + THE COMPONENT POCKETS + THE BOARD PRESS-FIT. NO LOCATOR PILLARS; THE CAVITY FLOOR IS A FULL 1.00 EVERYWHERE.
   LOCATE THE RESIN BRACE (SEPARATE PART) VIA MATCHING Ø3.2 RECESSES. FLOOR STAYS A FULL 1.00 (THE (33,55) RECESS IS SLOTTED). MODELED IN THE STEP.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN, MODELED. CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); LIP = inner (cavity-side) lip edge; FRAME = proud back-frame bottom edges; BOSS = boss/spotface bottom edges (Detail B). BREAK ALL OTHER EXPOSED EDGES 0.10x45°. CONCAVE JUNCTION CORNERS (BOSS
6. FLOOR IS NOW A TRUE 1.00 (0.95 LOCAL UNDER THE U2 POCKET), AT THE ~1.0 Ti MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS.
   CAVITY 1.80 +-0.05 -> 1.75 WORST-CASE, MINUS WS17 1.70 MAX (DATASHEET CASE WS17 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. U2 RELIEF POCKET: 7.8 x 5.4 CENTERED (28.5, 37) ON THE CAVITY FLOOR, 0.05 DEEP (LOCAL FLOOR 0.95) - U2 (1.75) KEEPS 0.10 AIR. GENERAL FLOOR 1.00. MODELED IN THE STEP.
8. FRONT SUPPORT LIP IS ASYMMETRIC (SECTION A-A): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 CLEARING THE JP1/TP1 PADS, THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), AND THE C10/D10 CLUSTER; WIDENING TO 2.5 ONLY AT THE N/S ENDS. THE EXTERIOR BACK BORDER (PLAN) IS INDEPENDENT AND UNIFORM 2.0 ON ALL 4 SIDES.
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 4× M2×3 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.